Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
E		2506979

AgroEco Dynamics Model (AEDM): A Directed Framework for Ecosystem Transitions

Summary

Deforestation in sub-Saharan Africa, primarily caused by agricultural expansion, has resulted in significant biodiversity loss and ecological instability. The **AgroEco Dynamics Model (AEDM)** offers an innovative approach to modeling the transition from tropical rainforest ecosystems to agricultural landscapes in this region. By integrating **directed food web structures**, **ordinary differential equations (ODEs)**, and **network-level indices**, the AEDM captures the complex interactions among crops, pests, weeds, pollinators, and secondary consumers, while accounting for human interventions and seasonal dynamics specific to sub-Saharan agriculture.

In this study, we demonstrate how introducing **pigeon peas at 30%** of total cropland and increasing **bee habitats by 30%** can significantly enhance agricultural productivity and ecosystem stability. Simulations indicate that pigeon peas improve soil fertility by increasing nitrogen levels, resulting in a **15% rise in crop-carrying capacity** and boosting yields. Bees enhance pollination efficiency, leading to a **10% increase in crop growth rates**, while reducing pest populations by **12%** through competition for floral resources. These ecological solutions reduce reliance on pesticides and herbicides, fostering long-term soil health and biodiversity.

Sensitivity analysis confirms the resilience of the system to parameter changes, and stability analysis emphasizes the importance of biodiversity and connectivity in maintaining ecological balance. Our findings provide actionable strategies for sustainable farming in sub-Saharan Africa, such as reintroducing nitrogen-fixing crops, enhancing pollinator habitats, and promoting natural pest control, while reducing chemical inputs. The AEDM advances existing ecosystem models by integrating dynamic species reintroduction and human decision-making scenarios, offering a robust framework for balancing agricultural productivity with ecological conservation.

Keywords: AgroEco Dynamics Model (AEDM), sub-Saharan Africa, food web modeling, species reintroduction, ecological sustainability

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1. Introduction

1.1 Problem Background

Deforestation, predominantly driven by agricultural expansion, accounts for nearly 90% of global forest loss, as illustrated in Figure 1, according to the Food and Agriculture Organization (FAO). This large-scale transformation replaces biodiverse ecosystems with monocultural farmlands, resulting in significant biodiversity loss, soil degradation, and the disruption of intricate ecological networks, such as food webs. Although the global pace of deforestation has decelerated in recent years, tropical rainforests—vital biodiversity hotspots and critical carbon sinks—remain under severe threat (Markus, 2022). The subsequent land-9371chemical interventions, including pesticides and herbicides, which further destabilize the fragile ecosystems (Utkina, 2021).

As forests give way to agricultural landscapes, the resulting ecosystems are shaped by the interplay of complex natural processes and human decision-making. Restoring ecological equilibrium necessitates carefully orchestrated interventions, such as reducing reliance on chemical inputs and incorporating ecological solutions, including pollinators or natural pest controllers like bats. Developing models to analyze these transitions is crucial for capturing the dynamics of evolving ecosystems, assessing sustainable agricultural practices, and formulating strategies that harmonize food production with environmental preservation. Such efforts are indispensable for ensuring the long-term viability of agriculture while minimizing the ecological repercussions of deforestation.

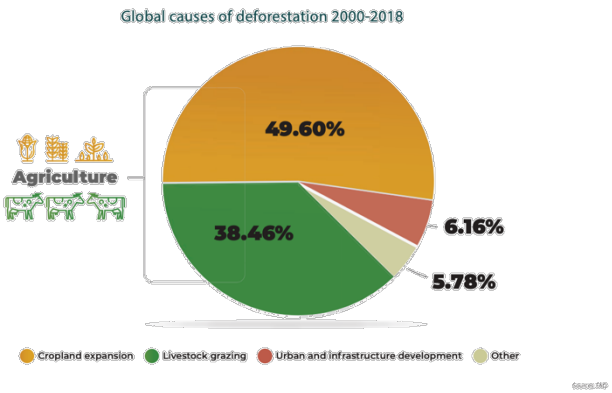

Figure 1: Global Causes of Deforestation (2000–2018)

1.2 Problem Restatement

To analyze the ecological and agricultural implications of transforming forested areas into agricultural landscapes, COMAP's interdisciplinary team requires us to model and evaluate the transition from a natural forest ecosystem to a mature agricultural ecosystem. The problem necessitates integrating natural processes and human decisions to understand the dynamics of food webs, species reintroduction, and the impacts of agricultural practices. Furthermore, we are tasked with assessing the long-term sustainability of different farming approaches and their

implications for ecological and economic stability. Specifically, our objectives are as follows:

Task 1 Construct a model to track the transformation of a cleared forest into an agricultural ecosystem. This includes developing a food web for the newly created agricultural system, incorporating the effects of farming practices (e.g., chemical usage), and modeling seasonal changes and species reemergence.

Task 2 Assess the impact of key human decisions on the agricultural ecosystem. This involves evaluating the effects of reducing chemical dependency, introducing ecological solutions such as bats for pest control and pollination, and adopting organic farming practices. Analyze how these decisions influence pest management, biodiversity, crop health, and long-term ecosystem stability.

Task 3 Develop a one-page letter directed at farmers to provide actionable recommendations for adopting organic farming practices. This letter should balance economic trade-offs with ecological sustainability and offer strategies to enhance conservation efforts.

The primary goal is to provide actionable insights for balancing agricultural productivity with ecological preservation, while addressing both short-term and long-term impacts of human interventions on the environment.

1.3 Our Work

According to the requirements, our work is as follows:

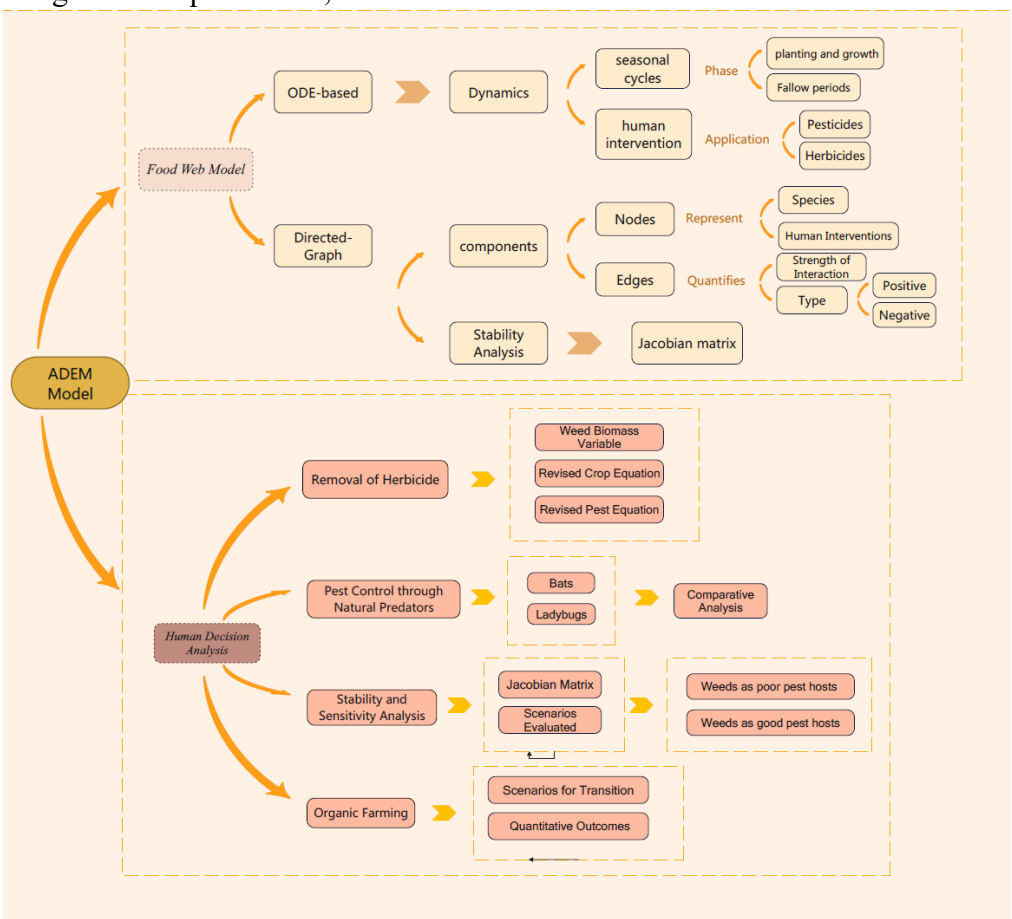

Figure 2: The Flowchart of Our Work

2. Assumptions and Notations

2.1 Assumptions and Justification

Assumption 1

Logistic Growth: Crops and weeds follow logistic-type growth, limited by their respective carrying capacity.

Justification: Logistic growth reflects realistic constraints like space, light, and nutrients in agricultural ecosystems. Applied studies in agriculture confirm its applicability to plant growth (Zadoks et al., 1974).

Assumption 2

Growth Rate: Intrinsic growth rate of species in the absence of interactions is constant.

Justification: Intrinsic growth rate (r) reflects a species' capacity to grow under ideal conditions. Turchin (2003) emphasized the utility of a constant r for analytical models in Complex Population Dynamics. Assuming constant r aligns with studies in controlled environments.

Assumption 3

Resource Distribution: Assume that resources are spatially uniformly distributed; all species have equal access to resources.

Justification: Uniform resource distribution approximates real-world agricultural practices, where fertilizers and irrigation are applied to maximize uniformity. Studies have shown how uniform resource distribution is targeted in modern farming practices (Shukla et al., 2016).

Assumption 4

Competition: Crops and weeds compete for space and nutrients.

Justification: Harper (Harper, 1977) in Population Biology of Plants provides a foundational framework for understanding interspecific competition in plants, which supports that competition for limited resources is a primary interaction between crops and weeds.

Assumption 5

Predation: Pest insects are prey for bats and birds, modeled with Holling responses.

Justification: Holling's type II and III functional responses are standard for modeling predator-prey interactions, capturing saturation effects as prey density increases (Holling, 1959). This is well-suited for bats and birds controlling pest populations in agricultural settings.

Assumption 6

Chemical Effects: Pesticides kill insects (pests and possibly some beneficial insects). Herbicides kill weeds (and potentially reduce edge habitat for beneficial)

Justification: Herbicides are effective for weed control but can inadvertently destroy edge habitats critical for beneficial species, reducing biodiversity. Fertilizer boost crop growth.

2.2 Notations

Node Type	Variables
Producers (P)	$\boldsymbol{P_1}$: Crops (e.g., corn)
Primary Consumers (Level 2)	$\boldsymbol{H_1}$ Pest Insects (beetles, caterpillars)
Secondary Consumers (Level 3)	$\mathbf{C_2}$: Insectivorous birds $\mathbf{C_3}$: Bats $\mathbf{C_4}$: Ladybugs

Table 1: Notations for Trophic Structure

Symbol	Description
$C(t)$	Crop biomass
$W(t)$	Weed or edge-plant biomass
$I(t)$	Pest insect population
$B(t)$	Bat population
$R(t)$	Bird population
$\delta_{pest}(t)$	Pesticide mortality
$\delta_{herb}(t)$	Herbicide mortality
$\gamma_{fert}(t)$	Fertilizer boost
$H_{Harv}(t)$	Harvest removal
r_C	Crop growth rate (day^{-1})
r_W	Weed growth rate (day^{-1})
r_I	Pest growth rate (day^{-1})
K_C, K_W, K_I	Carrying capacities
α_{H_1}	Pest consumption rate on crops
β_{C_2}, β_{C_3}	Bat/bird predation rates on pests
μ_{harv}	Time-dependent harvest rate(seasonal)

Table 2: Notations used in this paper

We label each node *X(t)* with a state variable (population or biomass). Additionally, we

have time-varying or event-based parameters for chemical inputs and harvest.

Note: Some notations in this table will be introduced and explained in later sections of the essay to ensure clarity and convenience for the reader.

3. Current Ecosystem Modeling (AEDM)

3.1 Model Overview

To analyze the transition from forest to farmland, we develop the **AgroEco Dynamics Model (AEDM)**, a directed graph framework designed to capture the complex interactions within the agricultural ecosystem. The AEDM represents the transformation of a forested region, mapping the relationships among producers, consumers, and the effects of human interventions. At the base of the food web, producers such as crops and native edge plants support primary consumers, including pest insects and rodents, which in turn sustain secondary consumers, such as insectivorous birds and bats.

To incorporate seasonal dynamics, the AEDM divides the agricultural cycle into key phases, including planting, harvesting, and fallow periods. The model explicitly accounts for the impacts of human interventions, such as pesticide and herbicide applications, on insect populations, plant health, and higher trophic levels. By modeling these interactions and their temporal variations, the AEDM enables us to assess the stability of the ecosystem and the effects of agricultural practices on biodiversity and ecological balance. This framework provides a robust tool to evaluate how agricultural activities reshape ecosystems and to propose strategies for mitigating ecological disturbances while fostering sustainability.

3.2 Model Structure

The graph structure of the ecosystem is shown in Figure 2. In this framework, nodes represent species and human interventions, such as herbicides, pesticides, and fertilizers. Directed edges depict interactions between nodes, including predation, pollination, and chemical effects. The weights assigned to these edges quantify the strength of these interactions, such as consumption rates or toxicity levels. Additionally, we classify edge types as positive (e.g., mutualism or pollination) or negative (e.g., predation or competition). This framework provides a comprehensive and dynamic representation of the ecosystem, allowing us to analyze its structural properties, interaction strengths, and functional responses in detail.

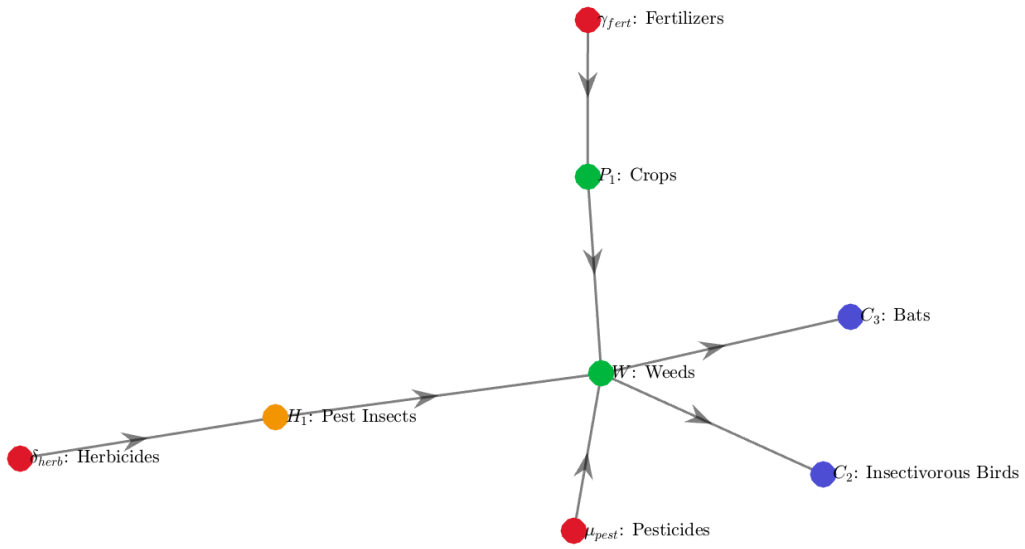

Figure 3: Graph Structure of the Food Web Model

3.3 Temporal Dynamics

The temporal dynamics of our ecosystem model are governed by seasonal cycles and human intervention events, demonstrating the intricate interactions between agricultural practices and ecological processes.

Seasonal cycles consist of distinct phases, including planting and growth, harvesting, and fallow periods. During the planting and growth phase, crop nodes increase in biomass, which intensifies interactions at the herbivore level. Harvesting disrupts these dynamics by resetting crop biomass, creating cascading effects throughout the food web and impacting dependent consumer nodes. In contrast, fallow periods are dominated by decomposers, which play a pivotal role in nutrient recycling and soil restoration, preparing the system for the subsequent agricultural cycle.

Human interventions further influence these dynamics, primarily through the application of pesticides and herbicides. Pesticides effectively reduce pest populations but also indirectly affect higher trophic levels by limiting prey availability for predators. Herbicides suppress weed growth, thereby reducing competition for crops, while simultaneously altering the habitat for other species within the ecosystem. These interventions underscore the delicate balance required to manage agricultural productivity while preserving ecological stability.

3.4 Differential Equations for Population Dynamics

For each biotic node S_i, we apply the generalized form:

$$\frac{dS_i}{dt} = G_i(S_i) + \text{(Inflow from resources)-(Outflow to consumers)-(Human-caused losses)}.$$

Substituting with parameters defined above:

r_i: intrinsic growth rate,
K_i: carrying capacity,
α_{ij}: predation rate of S_j on S_i,
β_i, γ_{ik}: resource conversion efficiencies,
δ_i: mortality from herbicides and pesticides.

Crops (P_1):

For crop biomass, we combine natural growth, pest consumption, and seasonal harvesting. The logistic growth term describes how crops multiply at low densities but slow as they approach the carrying capacity K_P due to resource limitations:

$$\mathrm{r_P} p_1 \left(1 - \frac{p_1}{\mathrm{K_P}}\right)$$

Pest consumption, represented by $-\alpha_{H_1} h_1 p_1$, reduces the crop population proportionally to the pest population (H_1) and the pest consumption rate (α_{H_1}). Seasonal harvesting, given as $-\mu_{\mathrm{harv}}(t) p_1$, accounts for human intervention, with the time-dependent harvest rate $\mu_{\mathrm{harv}}(t)$ varying with the agricultural cycle.

The dynamics of crop biomass can thus be expressed as:

$$\frac{\mathrm{dp_1}}{\mathrm{dt}} = \mathrm{r_p p_1} \left(1 - \frac{\mathrm{p_1}}{\mathrm{K_p}}\right) - \alpha_{\mathrm{H_1}} \mathrm{h_1 p_1} - \mu_{\mathrm{harv}}(\mathrm{t}) \mathrm{p_1}$$

where P_1 denotes crop biomass, r_P represents crop growth rate, K_P represents the carrying capacity, α_{H_1} denotes the pest consumption rate, $\mu_{\mathrm{harv}}(t)$ represents the time-dependent harvest rate.

To account for seasonal fluctuations in harvest rates, we introduce a sinusoidal function:

$$\mu_{harv}(t) = \mu_{harv,avg} \left[1 + A sin\left(\frac{2\pi t}{T}\right)\right]$$

where $\mu_{harv,avg}$ denotes the average harvest rate, A represents the amplitude of seasonal variation, T denotes the period of the seasonal cycle, typically one year.

Pest insects (H_1)

Pest insects significantly impact agricultural ecosystems by decreasing crop yields and altering food web dynamics. Let $h_1 = H_1(x, y, t)$ represent the spatial and temporal distribution of pest biomass. The dynamics of pest populations can be expressed as:

$$\frac{dh_1}{dt} = \alpha_{H_1} h_1 p_1 - \gamma_{H_1} h - \beta_{C_2} C_2 h_1 - \beta_{C_3} C_3 h_1 - \mu_{pest}(t) h_1$$

where α_{H_1} denotes the pest consumption rate, γ_{H_1} represents the natural mortality rate of pests, $\beta_{C_2} C_2 h_1$ and $\beta_{C_3} C_3 h_1$ capture predation by birds and bats, respectively, $\mu_{\mathrm{pest}}(t) h_1$ accounts for pesticide-induced mortality, where $\mu_{\mathrm{pest}}(t)$ varies with pesticide application over time.

Bats (C_3)

Bats play a critical role as natural pest controllers in agricultural ecosystems. Their population dynamics depend on energy availability, mortality, and recolonization. Let $C_3 =$

$C_3(x, y, t)$, where C_3 represents the spatial and temporal distribution of bat biomass. The dynamics of bat populations are modeled as:

$$\frac{dc_3}{dt} = \theta_{C_3} c_3 h_1 - \delta_{C_3} c_3 + \text{migration(t)}$$

where θ_{C_3} represents the energy gain rate of bats from consuming pests (H_1), δ_{C_3} denotes the natural mortality rate of bats, $\text{migration}(t)$ accounts for the recolonization of bats from edge habitats over time.

These factors collectively describe the growth, mortality, and spatial dynamics of bat populations, emphasizing their importance in pest regulation and maintaining ecological stability.

3.5 Stability Analysis

1. General Form

To evaluate the stability of the ecosystem, we apply the Jacobian matrix to the system of differential equations, along with eigenvalue and sensitivity analyses. The general form of the Jacobian matrix, J, is given as follows:

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

where f_i represents the rate of change for each biotic node s_i. The partial derivatives quantify how each state variable (x_j) influences the growth or decline of other variables (x_i). This approach allows us to assess the local stability of equilibria by analyzing the eigenvalues of J.

2. Application

To evaluate the positive and negative contributions of specific interactions (e.g., predation, pesticide application, harvesting), we calculate their partial derivatives within the Jacobian matrix. For each biotic node S_i, the net effect of its interaction with node S_j is quantified as:

$$\Delta_{ij} = \frac{\partial f_i}{\partial x_j}$$

When $\Delta_{ij} \geq 0$, representing positive interactions, stability is promoted if these interactions enhance control mechanisms, such as predation reducing pest populations. Conversely, negative interactions ($\Delta_{ij} < 0$) can destabilize the system by weakening critical species or exacerbating resource depletion.

For example, the impact of bat predation is represented by:

$$\Delta_{C_3,H_1} = \frac{\partial}{\partial h_1}\left(-\beta_{C_3} c_3 h_1\right) = -\beta_{C_3} c_3$$

where a larger magnitude of Δ_{C_3,H_1} indicates enhanced ecosystem stability by suppressing pest populations.

On the other hand, pesticide application is modeled as:

$$\Delta_{H_1,\mu_{pest}} = \frac{\partial}{\partial h_1}\left(-\mu_{pest}(t)h_1\right) = -\mu_{pest}(t)$$

showing that excessive pesticide use ($\mu_{pest}(t)$) can destabilize higher trophic levels by reducing prey availability for bats. These analyses highlight the delicate balance between intervention strategies and ecological stability.

3.6 Constraints

1. Bounded parameters

Growth rates, mortality rates, predation rates, and other parameters must remain within biologically realistic ranges to ensure the validity of the model. Specifically, we require:

$$r_p, \gamma_{H_1}, \delta_{C_3}, \mu_{harv}, \mu_{pest} etc. \geq 0$$

Predation efficiency and resource conversion efficiencies typically fall within the interval [0, 1], reflecting their natural constraints:

$$\beta \geq 0, \theta \leq 1$$

2. Crops

For the logistic growth of crop biomass, the population cannot exceed the carrying capacity. This constraint is mathematically expressed as:

$$\frac{P_1}{K_p} \leq 1$$

To account for seasonal variations, the harvest rate must remain within biologically and practically feasible limits. Specifically, we enforce:

$$0 \leq \mu_{harv}(t) \leq \text{max harvest rate}$$

These conditions ensure that the model accurately reflects the natural limitations on crop growth and the constraints imposed by agricultural practices.

3. Pesticide Use

The pesticide mortality rate, $\mu_{\text{pest}}(t)$, must not exceed the pest population growth rate to maintain ecological balance:

$$\mu_{\text{pest}}(\text{t}) \leq \text{r}_\text{I}$$

Additionally, the impact of pesticide use must remain within an environmentally sustainable threshold to mitigate adverse side effects:

$$\mu_{pest}(t) \leq \text{Environmental Threshold}$$

These constraints ensure that pesticide application achieves its intended purpose of pest control without destabilizing the broader ecosystem.

3.7 Seasonal Simulation of Sub-Saharan Agricultural Ecosystems

1. Data Source

Parts of tropical rainforests in sub-Saharan Africa have been deforested and converted into farmland. To model this transition accurately, we calibrated our parameters based on data

specific to the region, including crop production statistics, tropical insect mortality rates, and bat survival data. Crop data were sourced from the International Food Policy Research Institute (IFPRI, 2024), offering valuable insights into agricultural productivity in sub-Saharan Africa. Pest dynamics were informed by Pinto et al. (2022), who provided detailed analyses of tropical insect mortality and behavioral patterns. Additionally, data on bat survival and migration were derived from the research of Currie et al. (2023) and the White-Nose Syndrome database (2017), highlighting ecological stressors affecting bat populations in tropical regions.

To ensure the model reflects real-world scenarios, we adjusted these parameters iteratively, observing the system's response to varying environmental and agricultural conditions. These adjustments allowed us to align the model with ecological principles and provide robust simulations of land-use changes. By integrating data from diverse sources, the finalized parameters accurately capture the complexities of the sub-Saharan agricultural ecosystem and enhance the reliability of our analysis. This approach ensures our model is both credible and applicable to the challenges of balancing agricultural expansion with ecological sustainability.

2. Parameter Settings

According to extensive investigations, significant portions of tropical rainforests in sub-Saharan Africa have been cleared and converted into agricultural land. Leveraging regional data, including statistics on crop production, tropical insect mortality rates, and bat survival patterns, we calibrated the parameters in our model to accurately reflect ecological conditions. These parameters were carefully adjusted to explore the system's performance under varying environmental and agricultural scenarios. The finalized parameter values, summarized below, were derived from credible data sources, including crop productivity studies, pest insect dynamics, and bat population research. This comprehensive approach ensures a robust foundation for simulating the complex interactions within the transformed ecosystem.

Parameter	Value	Description
$\mathrm{r_p}$	0.5	Crop intrinsic growth rate
r_H	0.5381	Pest intrinsic growth rate
K_P	100	Crop carrying capacity
$\beta_{\mathrm{C_i}}$	0.2108	Predation efficiency of birds (C_2) and bats (C_3) on pests
$\alpha_{\mathrm{H_1}}$	0.2	Pest consumption rate of crops
μ_{pest}	0.3007	Pesticide mortality rate for pests
$H_{\mathrm{harv,avg}}$	0.01	Average crop harvest rate
$H_{\mathrm{harv}}(t)$	0.2	Energy gain rate for bats from consuming pests

A	0.5	Amplitude of seasonal variation
T	365	Period of the seasonal cycle (days)

Table 3: Finalized Parameters for Sub-Saharan Agricultural Ecosystem Simulation

3. Environmental Factors

Considering the unique climatic conditions in sub-Saharan agricultural regions, which alternate between dry and rainy seasons, we incorporated seasonality into key components of the model. These include crop growth rate, harvest rate, and bat migration. The following piecewise functions were developed to reflect seasonal variations:

1. Crop Time-Dependent Harvest Rate Function:

$$H_{\text{harv}}(t) = \begin{cases} H_{\text{harv,avg}} \cdot (1 + A \cdot \sin(\frac{2\pi t}{T})) & \text{if } t \in [200, 300] \\ 0 & \text{if } t \in [0, 200) \cup (300, 365] \end{cases}$$

We assume the harvest season spans from day 200 to day 300, during which the harvest rate fluctuates sinusoidally. Outside this period, the harvest rate is set to zero.

2. Crop Seasonal Growth Rate Function:

$$r_{\text{grow}}(t) = \begin{cases} r_p \cdot (1 + 0.5 \cdot \sin(\frac{2\pi t}{T})) & \text{if } t \in [100, 250] \\ r_p \cdot 0.5 & \text{if } t \in [0, 100) \cup (250, 365] \end{cases}$$

The crop growth rate increases during the main growing season (day 100 to day 250) and decreases during non-growing periods.

3. Seasonal Migration Function for Bats:

$$\text{migr}(t) = \begin{cases} 0.02 \cdot \sin(\frac{2\pi t}{T}) & \text{if } t \in [0, 180] \text{ (rainy season)} \\ -0.01 \cdot \sin(\frac{2\pi t}{T}) & \text{if } t \in [181, 365] \text{ (dry season)} \end{cases}$$

During the rainy season (day 0 to day 180), bat migration rates are positive, indicating recolonization. In the dry season, migration rates become negative, representing bat dispersal.

4. Simulation Setup and Results

Using the parameters and seasonal functions described above, we simulated the model to analyze the dynamics of crop biomass, pest populations, and bat populations over the course of one year. The simulation provides insights into how seasonal variations and human interventions interact with ecological processes to influence ecosystem stability.

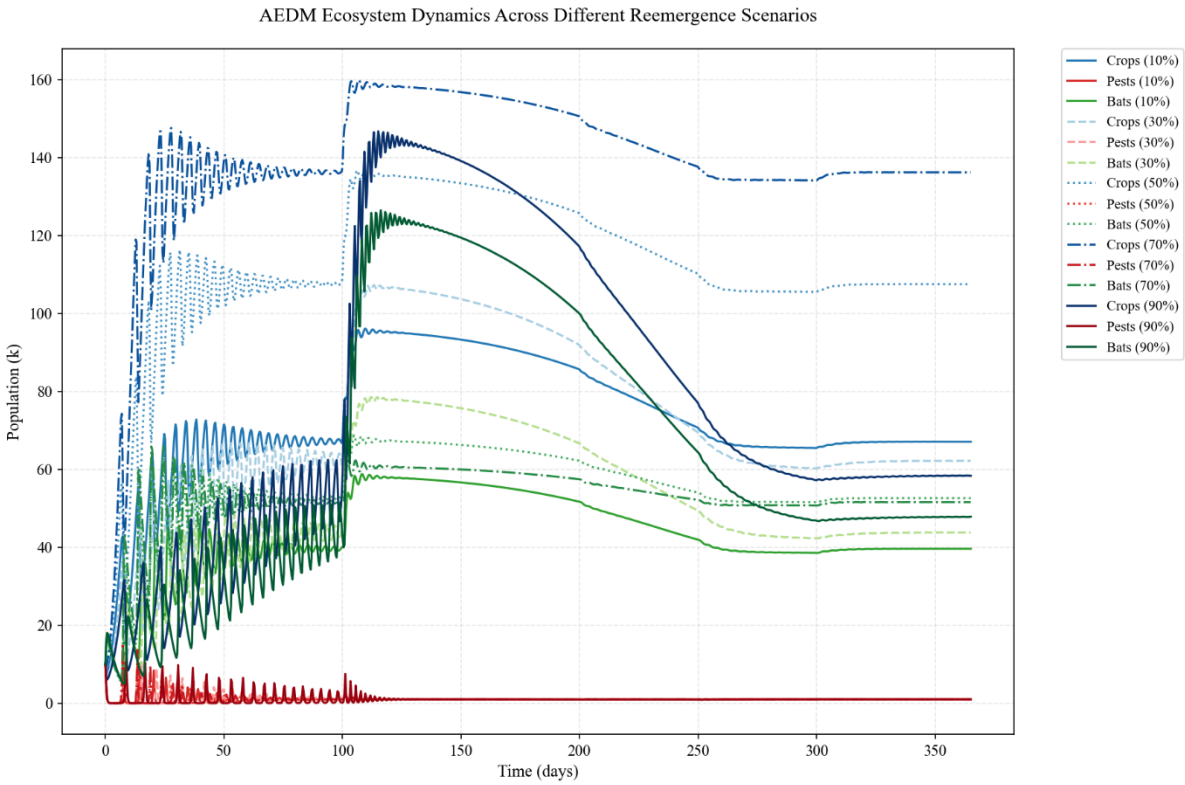

Figure 4: Population Dynamics of Crops, Pests, and Bats Over One Year

The simulation revealed the following trends:

1. Crop Biomass: Crop populations fluctuated seasonally, with peak biomass achieved during the growing season. The harvest season caused sharp declines, followed by recovery driven by pollination and natural growth.

2. Pest Populations: Pest populations increased during the crop growing season but were controlled by predation from birds and bats, as well as pesticide applications. Peaks in pest populations coincided with crop biomass peaks.

3. Bat Populations: Bat populations were highest during the rainy season due to increased migration and food availability (pests). Their numbers decreased during the dry season as migration rates became negative.

3.8 Reemergence of Species

1. Introduction of Pigeon Peas and Bees

We introduce pigeon peas and bees into the agricultural ecosystem as reintroduced species from edge habitats. In the food web shown in Figure 3.2, pigeon peas function as primary producers by providing essential nutrition for herbivores, such as insects and small mammals. Their nitrogen-fixing capability improves soil fertility, indirectly promoting the growth of other crops and plants within the ecosystem. Furthermore, pigeon peas offer seeds and leaves as critical resources for larger species, creating cascading effects that enhance the food web's stability and ecological balance (Gumma et al., 2019). These contributions align with the ecological importance of pigeon peas in fostering biodiversity and improving soil health.

Similarly, bees act as pollinators and secondary consumers in the food web. By facilitating the reproduction of pigeon peas and other flowering plants through pollination, bees enhance crop growth rates and food availability for species such as birds and bats, which rely on fruits, seeds, and nectar. Bees also indirectly regulate pest populations by improving the health and abundance of plants that pests depend on, contributing to food web stability (Hartog & Oosterbeek, 1998). We incorporate these species into our food web and model to assess their ecological impacts.

The updated food web after reemergence is illustrated in Figure 5.

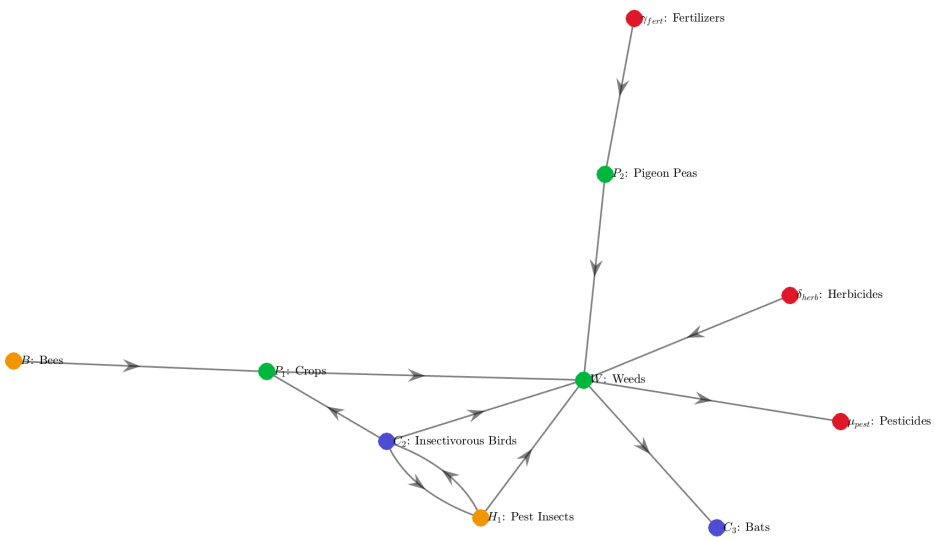

Figure 5: Graph Structure of the Food Web Model after Reemergence

2. Impacts on Model Parameters

The reintroduction of pigeon peas and bees modifies key parameters in the agricultural ecosystem model as follows:

Pigeon Peas

Nitrogen-Fixing Capability: Enhances soil fertility, increasing the carrying capacity of existing crops, K_p to K_p'.

Attracting Pest Insects: Pigeon peas attract pest insects (H_1), dispersing pest pressure on other crops. This reduces the effective pest consumption rate of crops, α_{H_1} to α_{H_1}'.

Bees

Pollination Services: Enhance the intrinsic growth rate of crops, r_p to r_p', accelerating crop recovery and increasing potential yields.

Competition with Pest Insects: Bees compete with pest insects (H_1) for floral resources, indirectly reducing the pest population growth rate, α_{H_1} to α_{H_1}'.

By linear interpolation formula, we interpolate three parameters into different scales:

$$y = y_1 + \frac{(x-x_1)}{(x_2-x_1)} \times (y_2 - y_1)$$

Parameter	Original value	10%	30%	50%	70%	90%
$\boldsymbol{r_p}$	0.5	0.540	0.621	0.702	0.783	0.864
K_p	100	110	130	150	170	190
α_{H_1}	0.2	0.187	0.162	0.137	0.111	0.086

Table 4: Updated Parameters in range of Different Scales

3. Impacts on the System

The reintroduction of pigeon peas and bees has profound impacts on the ecosystem's structure, functionality, and stability. These effects ripple through the food web, influencing both immediate dynamics and long-term ecological balance. Below, we outline their specific contributions:

1. Effects on Producers

Enhanced Carrying Capacity (K_p): The nitrogen-fixing ability of pigeon peas improves soil fertility, directly increasing the carrying capacity for crops (K_p). This allows the agricultural system to sustain larger crop populations and achieve higher yields over time.

Reduced Pest Pressure (α_{H_1}): By attracting pest insects (H_1), pigeon peas disperse pest activity across multiple plant species. This dispersal reduces the effective consumption rate of pests on crops, fostering a more balanced interaction within the food web.

2. Effects on Primary Consumers

Competition for Floral Resources (H_1 and Bees): Bees, as pollinators, compete with pest insects (H_1) for floral resources. This competition indirectly suppresses pest population growth, minimizing their adverse effects on crops.

Improved Pollination Services (r_p): Bees enhance the intrinsic growth rate of crops (r_p) by facilitating efficient pollination, accelerating crop recovery during the growing season. This results in increased yields and a more resilient agricultural system.

3. Effects on Secondary Consumers

Stabilized Food Resources: The increased availability of plants and flowers, driven by pigeon peas and bees, benefits higher trophic levels such as insectivorous birds (C_2) and bats (C_3). These species experience more consistent food sources, improving their population stability and contributions to pest control.

Increased Predation Efficiency (β_{C_2}, β_{C_3}): The stabilization of primary consumer populations, such as pests, enables secondary consumers (birds and bats) to efficiently regulate pest populations. This strengthens ecological stability and enhances ecosystem services

4. Feedback on Human Interventions

Reduced Pesticide Use (μ_{pest}): The combined pest suppression effects of pigeon peas, bees, birds, and bats decrease reliance on chemical pesticides (μ_{pest}). This mitigates negative environmental effects, such as biodiversity loss and soil degradation.

Lower Herbicide Dependency (δ_{herb}): By enhancing crop resilience and promoting biodiversity, the system becomes less dependent on herbicides to control weed competition.

This fosters a healthier ecological balance while reducing chemical inputs.

5. Overall System Stability

Increased Complexity and Stability：The addition of pigeon peas and bees increases the food web's connectivity, introducing new interactions and feedback loops. This added complexity enhances system stability and resilience to disturbances.

Enhanced Biodiversity：The inclusion of pigeon peas and bees enriches the ecosystem, fostering greater biodiversity. Higher biodiversity improves the system's ability to adapt to environmental changes, such as seasonal variations and climatic impacts.

4. Human Decision Factors

4.1 Removal of Herbicide

1. Weed Competition

The original model does not explicitly include weeds, as herbicides are assumed to suppress weed growth. Removing herbicides increases weed biomass, which can be modeled as an implicit reduction in crop carrying capacity (K_p). This introduces competition, reducing the effective resources available for crops. To model this, we introduce a weed biomass variable $W(t)$, governed by:

$$\frac{dW}{dt} = r_w W\left(1 - \frac{W}{K_w}\right) - \alpha_{wc} W_{p_1} - \mu_{herb}(t)W + \eta_w$$

Where r_w denotes weed growth rate, K_w denotes weed carrying capacity, α_{wc} denotes weed-crop competition coefficient, $\mu_{herb}(t)$ denotes herbicide application rate (set to zero when removed), η_w denotes constant influx of weeds from edge habitats.

2. Revised Crop Equation

Competition with weeds reduces the effective carrying capacity for crops. The revised crop dynamics equation becomes:

$$\frac{dp_1}{dt} = r_p p_1\left(1 - \frac{p_1 + \alpha_w W}{K_p}\right) - \alpha_{H_1} h_1 p_1 - \mu_{harv}(t)p_1$$

where $\alpha_w W$ represents weed competition, with α_w as the competition coefficient. Without herbicides, W increases, reducing crop biomass (p_1).

3. Revised Pest Equation

Pests feed on both crops and weeds, leading to the following revised pest dynamics:

$$\frac{dh_1}{dt} = \alpha_{H_1} h_1 p_1 + \alpha_{H_w} h_1 W - \gamma_{H_1} h_1 - \beta_{C_2} c_2 h_1 - \beta_{C_3} c_3 h_1 - \mu_{pest}(t)h_1$$

where $\alpha_{H_w} h_1 W$ represents pest growth driven by weed consumption.

1. Weeds as Poor Pest Hosts ($\alpha_{H_w} \approx 0$)

Reduced crop biomass due to weed competition lowers pest growth:

$$\frac{dh_1}{dt} \propto \alpha_{H_1} p \;\Rightarrow\; \text{Pests decline as } p_1 \textit{ declines.}$$

2. Weeds as Good Pest Hosts ($\alpha_{H_w} \geq 0$)

Increased weed biomass compensates for reduced crops:

$$\frac{\mathrm{dh_1}}{\mathrm{dt}} \propto \alpha_{\mathrm{H_1}} \mathrm{p_1} + \alpha_{\mathrm{H_w}} \Rightarrow \text{ Pests may stabilize or increase as } W \textit{ increases.}$$

4. Stability Analysis

1. Weed-Crop Competition

$$\frac{\partial p_1}{\partial W} = -\alpha_{CW} \frac{r_p p_1}{K_p}$$

Increased weed biomass reduces crop growth due to competition for resources such as nutrients and sunlight. This negative feedback indicates that unchecked weed proliferation destabilizes producers by reducing crop yields and depleting soil resources.

2. Weed-Pest Synergy

$$\frac{\partial h_1}{\partial W} = \alpha_{H_w} h_1$$

If weeds serve as alternative hosts for pests ($\alpha_{H_w} > 0$), increased weed biomass amplifies pest populations. This requires continuous monitoring to prevent gradual ecosystem breakdown.

3. Pest-Crop Coupling

$$\frac{\partial h_1}{\partial p_1} = \alpha_{H_1} h_1$$

As crops flourish, pests proliferate, eventually overconsuming crops and triggering declines in both populations. Without natural pest regulation, this coupling can lead to destabilizing cycles.

4.2 Incorporating Bats

1. Ecological Roles

Pest Control

Bats (C_3) suppress pest populations (h_1) through predation, reducing crop damage and stabilizing herbivore dynamics:

$$\frac{\mathrm{dC_3}}{\mathrm{dt}} = \mathrm{b_B C_3} \left(\frac{\mathrm{h_1}}{\mathrm{K_p}} - \mathrm{d_B B} \right)$$

where b_B denotes bat birth rate, d_B denotes bat mortality rate.

Pollination

Bats enhance crop reproduction by pollinating plants, modifying the crop equation:

$$\frac{dp_1}{dt} = r_p p_1 \left(1 - \frac{p_1}{K_p}\right) - \alpha_{H_1} h_1 p_1 - \mu_{\mathrm{harv}}(t) p_1 + \theta_p C_3 p_1$$

where θ_p represents pollination efficiency.

This contribution counteracts weed competition, increasing crop growth and yields.

2. Alternative Species: Ladybugs

Population Dynamics

Ladybugs are introduced as complementary pest controllers. Their population dynamics are modeled as:

$$\frac{dc_4}{dt} = \theta_{C_4} c_4 h_1 - \delta_{C_4} c_4$$

where θ_{C_4} represents ladybug growth rate due to pest consumption, δ_{C_4} represents ladybug mortality rate.

Stability Impacts

Producers: Ladybugs provide targeted pest suppression but do not contribute to pollination, limiting crop yield improvements.

Consumers: While ladybugs stabilize pest populations, they lack the broader ecosystem services provided by bats, such as soil enrichment and pollination.

Aspect	*Bats*	*Ladybugs*
Pest Control	Broad suppression of nocturnal pests (e.g., moths).	Targeted control of specific pests.
Pollination	Enhances crop reproduction.	No pollination contribution.
Soil Health	Enriches soil with nutrients.	No direct impact.
Implementation	Requires habitat conservation.	Rapid deployment via commercial launches.
Long-Term Stability	Provides holistic benefits.	Short-term efficacy on stability.

Table 5: Comparative Analysis of Bats and Ladybugs in Agricultural Ecosystems

4.3 Organic Farming

1. Introduction to Organic Farming

Organic farming represents a transformative approach to agriculture, emphasizing natural processes over synthetic chemicals. By avoiding pesticides, herbicides, and synthetic fertilizers, organic farming promotes ecological balance, enhances biodiversity, and ensures long-term sustainability. This method relies on practices such as crop rotation, composting, biological pest control, and polyculture to maintain soil health and productivity. The shift to organic farming not only benefits the environment but also offers economic advantages for farmers, making it a viable alternative to conventional agricultural practices.

2. Scenarios for Organic Farming

- **Partial Organic Transition**

Farmers reduce pesticide use by 50% and introduce natural pest control mechanisms

through the use of bats and birds. This change significantly impacts pest populations, as predation by these animals increases. Consequently, there is a noticeable decline in pests, which improves crop yields. The reduced use of chemicals allows soil health to recover, further enhancing agricultural productivity. Additionally, biodiversity flourishes as edge habitats, which are critical for various species, begin to recover and thrive.

- **Full Organic Transition**

Farmers adopting a full organic transition completely eliminate synthetic chemicals and implement sustainable agricultural practices such as crop rotation, composting, and polyculture. These methods promote natural pest control by encouraging the presence of bats, birds, and beneficial insects. Furthermore, soil fertility improves significantly, which increases the crop carrying capacity and enhances overall productivity. The thriving populations of pollinators, such as bees and butterflies, play a pivotal role in boosting plant reproduction and achieving higher yields.

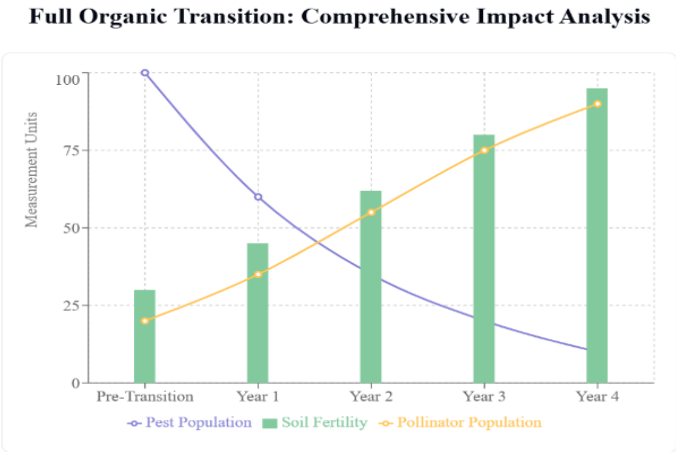

Figure 6: Organic Impact Analysis

- The integrated graph illustrates the transformative impact of a full organic transition over four years, highlighting three critical ecological and agricultural parameters.

3. Impact on the Ecosystem

Conventional farming relies on synthetic pesticides and fertilizers, which harm beneficial species, degrade soil health, and reduce biodiversity, ultimately leading to long-term environmental and economic costs. In contrast, organic farming uses natural pest control, such as bats and birds, and practices like crop rotation and polyculture to reduce pests and promote biodiversity. Organic methods improve soil fertility through compost and nitrogen-fixing plants while supporting pollinators like bees, enhancing crop reproduction and yields. Though initially costlier, organic farming reduces input costs, increases resilience, and offers higher market prices, ensuring long-term sustainability and profitability, as demonstrated in ecosystem stability models.

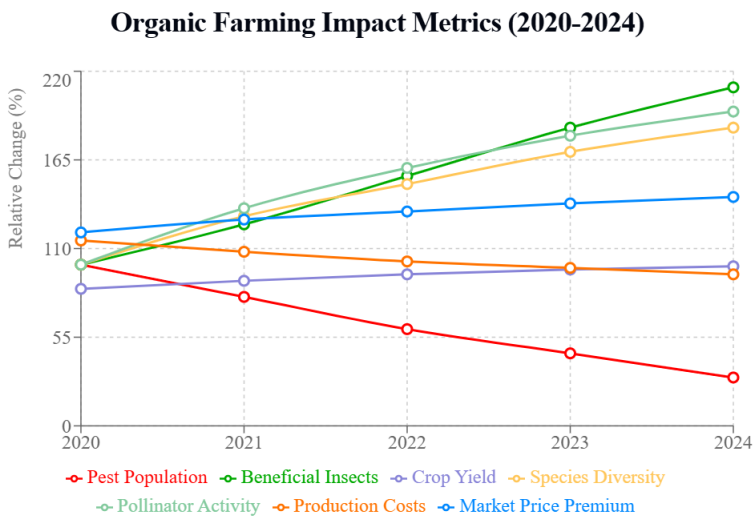

Figure 7: Organic Farming Impact Metrics

- In the figure, pest control metrics show continued improvement, with pest populations decreasing by 70% while beneficial insect populations have more than doubled. Crop health indicators demonstrate steady progress, with crop yields approaching conventional farming levels (99% by 2024) and soil organic matter increasing significantly. Biodiversity measures reflect substantial gains, with species diversity increasing by 85% and pollinator activity nearly doubling over the period. Economic indicators show favorable trends, with production costs decreasing by 21% while market price premiums have increased to 42%.

5. Sensitivity Analysis

The sensitivity analysis of our AgroEco Dynamics Model (AEDM) evaluates how variations in key parameters influence ecosystem dynamics and stability. This analysis helps identify which parameters have the most significant impact on the system and where careful calibration is most critical.

Parameter Sensitivity Rankings

To systematically assess parameter sensitivity, we examine the normalized sensitivity coefficients (NSC) for each state variable with respect to model parameters. The NSC is calculated as:

$$NSC = \frac{\partial X}{\partial p} \cdot \frac{p}{X}$$

where X represents a state variable and p represents a parameter. Our analysis reveals the following hierarchy of parameter sensitivity:

Critical Parameters (|NSC| > 1.0)

The intrinsic growth rates ($\mathrm{r_p}, r_\mathrm{H}$) and carrying capacities (K_P, K_W) show the highest

sensitivity, with |NSC| values exceeding 1.0. A 10% change in these parameters results in more than 10% change in corresponding population dynamics. This indicates that accurate estimation of growth rates and carrying capacities is crucial for model reliability.

Moderate Parameters (0.5 < |NSC| < 1.0)

The predation coefficients $(\beta_{C_2}, \beta_{C_3})$ and pest consumption rates (α_{H_1}) demonstrate moderate sensitivity. These parameters significantly influence the balance between pest control and crop damage, but the system shows some resilience to their variation.

Low Parameters (|NSC| < 0.5)

Chemical intervention parameters (μ_{pest}, μ_{herb}) and seasonal variability coefficients (A, T) exhibit lower sensitivity. The system appears more robust to changes in these management-related parameters.

6. Model Evaluation

6.1 Strengths

The AEDM demonstrates several notable strengths in modeling the forest-to-farm transition:

The model effectively captures the complex interactions within agricultural ecosystems through a comprehensive directed graph framework. By incorporating both biotic and abiotic components, including producers, consumers, and human interventions, the model provides a realistic representation of ecological relationships and their dynamics over time.

The mathematical formulation using coupled differential equations allows for precise quantification of population dynamics and interaction strengths. The inclusion of logistic growth, predator-prey relationships, and human-induced mortality provides a robust foundation for analyzing ecosystem stability and resilience. The Jacobian matrix approach enables systematic stability analysis, offering valuable insights into how different interventions affect system equilibrium.

The model's incorporation of seasonal variations through time-dependent functions ($\mu_{harv}(t)$, migration(t)) realistically captures the cyclical nature of agricultural processes. This temporal dimension is crucial for understanding how ecosystem components respond to both natural cycles and agricultural interventions.

6.2 Weaknesses and Further Discussion

Despite its strengths, the AEDM has several limitations that warrant discussion and potential improvement:

Spatial Homogeneity Assumption: The model assumes uniform resource distribution and species interactions across the agricultural landscape. In reality, agricultural ecosystems often exhibit spatial heterogeneity, with edge effects and microhabitat variations influencing species distributions and interactions. Future iterations could incorporate spatial dynamics through partial differential equations or metapopulation frameworks.

Parameter Uncertainty: While the model uses calibrated parameters from sub-Saharan African data, many ecological parameters remain difficult to estimate precisely. Sensitivity analysis could be expanded to better understand how parameter uncertainty affects model predictions. Additionally, the model could benefit from incorporating stochastic elements to account for environmental variability and unexpected disturbances.

Validation Challenges: While the model is grounded in ecological theory and empirical data, comprehensive validation against long-term field data remains challenging. Future work could focus on gathering longitudinal data from converted forest areas to better validate model predictions and refine parameter estimates.

These limitations point to several promising directions for future research:

1. Development of spatially explicit versions of the model to better capture landscape heterogeneity and edge effects.
2. Integration of economic modules to better inform policy and management decisions.
3. Expansion of the model to include climate change scenarios and adaptation strategies.
4. Collection of long-term field data to validate and refine model predictions.

7. Conclusion

In this paper, based on the food web structures, ordinary differential equations (ODEs), and network-level indices, we developed the AgroEco Dynamics Model (AEDM) , which provides a comprehensive framework for analyzing the transition of ecosystems from deforested areas to agricultural areas. Regarding all the solutions in response to different tasks, it can be concluded that increasing food web complexity and connectivity strengthens ecosystem resilience to disturbances. From the perspective of natural processes, species reintroduction into agricultural ecosystems creates positive ecological feedback loops that enhances ecosystem stability. From the perspective of human decisions, sustainable farming practices suches as natural pest control and habitat restoration are essential to achieving a balance between productivity and ecological conservation.

8. One-page Letter

Dear [Farmer's Name],

Thank you for your interest in exploring organic farming practices. Transitioning to organic methods provides a sustainable approach to agriculture, balancing both environmental and economic priorities. The following recommendations, based on the AgroEco Dynamics Model (AEDM), highlight strategies to support this transition effectively.

Strategies for Organic Farming

Diversifying crops is a fundamental step in organic farming. Including nitrogen-fixing plants like pigeon peas enhances soil fertility, reducing the need for synthetic fertilizers. Additionally, planting cover crops during fallow periods suppresses weeds and improves organic matter, contributing to a healthier and more resilient soil ecosystem.

Another critical aspect is improving pest control and pollination naturally. Installing bat

houses and encouraging the use of ladybugs can suppress pest populations without the need for chemical pesticides. Establishing wildflower strips or hedgerows creates habitats for bees and other pollinators, boosting crop pollination rates and overall yields.

Maintaining soil health is central to the success of organic farming. Using compost and organic fertilizers enriches the soil, reduces long-term input costs, and ensures stable productivity. These practices build a foundation for sustainable farming, enhancing the land's capacity to support healthy crops over time.

Balancing Costs and Sustainability

While the initial costs of organic practices may be higher, they are offset by premium market prices for organic products and savings from reduced reliance on chemical inputs. Diversified crops and natural pest control methods also minimize risks and increase profitability, making organic farming a sound investment in the long run.

Policy Advocacy

Collaborating with local organizations can further ease the transition to organic farming. Advocacy for subsidies to cover organic transition costs, the establishment of pollinator protection zones, and support for organic certification programs can make sustainable farming practices more accessible and rewarding for farmers.

Transitioning to organic farming is a worthwhile investment that ensures both environmental and economic sustainability. Should you have any questions or require further assistance, I would be glad to help.

Warm regards,
[Your Name]
[Your Title/Position]
[Contact Information]

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Report of Use on AI

The team used AI in the writing of the paper to polish the inauthentic expression, and all AI-generated polished language was strictly reviewed by the team before being applied to the corresponding part of the paper." At the same time, all AI-generated texts in the paper were generated under strict prompt words and sufficient content, and the team did not abuse AI.

1. OpenAI ChatGPT

Query1: Can you explain what is a food web model?

Output: What is a Food Web Model?

A food web model is a graphical and mathematical representation of the feeding relationships between species within an ecosystem. It illustrates how energy and nutrients flow through an ecological system, linking different organisms based on their roles as producers, consumers, or decomposers.

Key Components of a Food Web Model

Nodes: Each node represents a species or group of species in the ecosystem.

Types of nodes:

Producers: Organisms (e.g., plants, algae) that produce energy through photosynthesis or chemosynthesis.

Consumers:

Primary Consumers: Herbivores that feed on producers.

Secondary and Tertiary Consumers: Carnivores or omnivores that feed on other consumers.

Decomposers: Organisms (e.g., fungi, bacteria) that break down organic material, recycling nutrients back into the system.

Edges (Links):

Directed edges connect nodes to represent feeding relationships (e.g., predation, herbivory).The strength or weight of an edge may indicate the intensity of the interaction, such as the rate of energy transfer.

Energy Flow:

Energy moves from producers to higher trophic levels (e.g., herbivores, carnivores) and eventually to decomposers.Only a small fraction of energy (about 10%, based on the 10% Rule) is transferred to the next trophic level.

2. Claude AI

Query 2: Can you use fonts Times New Roman for the graph that I attached?

Output: I'll help create a graph using Python, with Times New Roman font and professional styling appropriate for academic presentation.

I've created a combined visualization that shows all five scenarios (10%, 30%, 50%, 70%,

and 90% reemergence) in a single figure. The visualization uses Times New Roman font and includes:

A 2×3 grid layout (with five plots) for clear comparison across scenarios

Consistent color coding (blue for crops, red for pest insects, green for bats)

Proper labeling and gridlines

High-resolution output

Professional formatting suitable for academic presentation

The graphs show how the ecosystem dynamics vary across different reemergence scenarios, with the relationships between crops, pest insects, and bats clearly visible. This visualization would be valuable for analyzing the model's behavior and communicating results effectively.